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High-performance adhesives modified by demethylated lignin for use in extreme environments†

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Bisphenol A, a toxic chemical, is widely used in the preparation of epoxy resins. Therefore, the use of harmless biomass-based raw materials to replace bisphenol A in the preparation of epoxy resin has attracted extensive attention. A demethylated-lignin-based epoxy resin adhesive was prepared. After lignin was modified in a LiBr/HBr system, the phenolic hydroxyl group content of the obtained demethylation lignin (DL) increased to 6.19 mmol g^{-1}, and the tensile strength of the as-prepared DL-modified adhesive (–OH content of the DL = 6.19 mmol g^{-1}) increased to 62.50 MPa, which was much higher than the adhesive modified by raw lignin (38.89 MPa). Due to its excellent crosslinking density, the DL-modified adhesive retained more than 85% of its tensile shear strength performance after being soaked in water for 48 h or frozen at −25 °C for 48 h. Therefore, this adhesive has a wide range of application prospects in furniture and plywood manufacturing.

Introduction

Of all the wood adhesives produced, formaldehyde resin is the most widely used adhesive, accounting for more than 90% of the market share in 2013.[1,2] However, as early as the 1860s, people realized that formaldehyde was an air pollutant with strong irritation to the eyes and upper respiratory tract.[1,3] In the 1880s, relevant reports on the carcinogenic effects of formaldehyde began to appear.[1,4,5] Compared with formaldehyde resins, the preparation process of epoxy resins is formaldehyde free. Since the 1940s, epoxy resins have been greatly developed and widely used in industry. Epoxy resin comprises a three-dimensional network of a thermoset polymer, with excellent strength, adhesion ability, acid and alkali resistance, and thermal stability and is readily capable of being modified at the end groups to obtain different properties. Thus, epoxy resin is widely used in furniture manufacturing,[6] adhesives,[7] coatings,[8] plywood,[9] electronic component packaging,[10] and other high-performance materials for special purposes.[11] However, bisphenol A, a classic epoxy precursor, is used extensively in the manufacture of epoxy resins. As a chemically synthesized hormone, bisphenol A can interfere with human reproduction or the normal operation of the immune system and can harm human health.[12–15] Moreover, petroleum-based materials have brought about a series of environmental problems.[16] Aiming for environmental and health protection, cheap and sustainable biomass material alternatives for the preparation of epoxy resin adhesives would be of great significance. Qian and co-workers[17] prepared a novel epoxy resin, which exhibited excellent thermal properties (T_g of 114 °C) and mechanical properties (tensile strength of 60.2 MPa, toughness value of 185 MPa), using aminated-modified diphenolic acid and epichlorohydrin and by curing with succinic anhydride. Diphenolic acid was reported to be a promising sustainable candidate for replacing bisphenol A, and the novel cured epoxy resin displayed outstanding compatibility with cotton fibers. Zheng's group[18] synthesized a novel toughening benzoxazine using aminated-modified diphenolic acid and paraformaldehyde, and compared with commercial benzoxazine, its break at elongation was higher by 70.5% while its tensile strength reached 44.69 MPa.

Lignin is a natural polyphenol polymer with abundant reserves in nature and is also a by-product of the pulp and paper industry.[19,20] The rich phenol hydroxyl groups in lignin give it the potential to substitute bisphenol A to prepare lignin-based epoxy resins.[21,22] Nikafshar *et al.* investigated 13 lignin samples from different sources for the preparation of epoxidized lignin, which could replace 100% of bisphenol A in the formation of epoxy resins. The lignin-based epoxy resin prepared by lignin with a low molecular weight and high phenolic

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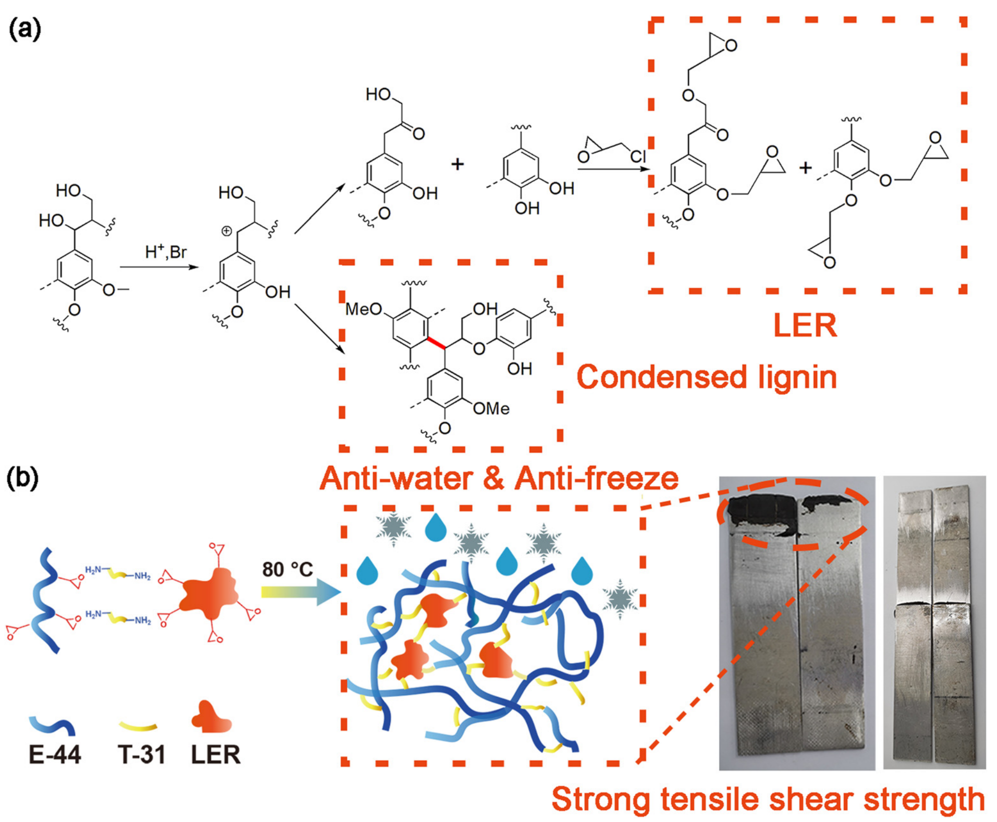
group content demonstrated a comparable performance to petroleum-based resins.[23] Hirose *et al.* prepared an epoxy resin with an extremely low glass-transition temperature (under 0 °C) using a Kraft lignin diglycidyl ether of polyethylene glycol.[24] However, the structure of industrial lignin is complicated and it has low chemical reactivity due to the generation of many condensed C–C linkages during the pulp making or pretreatment process.[25] In addition, the aggregation of lignin also reduces its compatibility with polymer compounds, resulting in phase separation inside the polymer materials, which leads to a stress concentration between the two phases and a decline in the performance of lignin-based polymer materials.[26,27] In order to improve the chemical reactivity of lignin and the thermal or mechanical properties of lignin-based polymer materials, the chemical modification of lignin is usually required.[28–32] Among these, the demethylation of lignin can transform the methoxyl group of lignin into a phenolic hydroxyl group to increase the content of phenolic hydroxyl groups of lignin, thus improving the crosslinking or modified reactivity of lignin with polymer compounds or specific functional groups contained in chemicals (*e.g.*, halogenated hydrocarbons, acyl halogenated, epoxy compounds). For example, the curing time of a lignin-based phenolic resin prepared by demethylated was shortened and its mechanical properties were improved.[33–35] Mei *et al.* successfully obtained acetic acid and polyphenols by the catalyzed depolymerization of Kraft lignin used $RhCl_3$ and LiI with additional H_2O and CO.[36] Wang *et al.* increased the content of phenolic groups through the phenolation of lignin, and proved that the demethylated-lignin-based sorbent agent had a strong adsorption for Pb(II).[37] While these lignin demethylation strategies are highly effective, most of the demethylation reagents are expensive, highly corrosive, and toxic, such as BBr_3 and AlI_3, which can efficiently demetheylate lignin and its model compounds at room temperature (Scheme 1).[38–40]

In our previous works, in order to improve the reactivity of lignin, we successfully increased the compatibility and hydrophobicity of lignin with commercially available epoxy resins through the alkaline hydrolysis of lignin[21] and grafted methacryloyl with lignin sulfonates,[41] and the prepared epoxy resin adhesives were able to resist boiled water and freezing. However, their relatively low tensile strength made it difficult for them to cope with the situations where furniture or plywood is subjected to great external stress, such as sudden hits, falls, and bumps in transit. Small water contact angle also made the adhesive have poor water resistance.

Herein, lignin was demethylated in a LiBr/HBr system (Scheme 1). The high concentration of LiBr, which has a much lower cost and toxicity among efficient demethylation reagents, catalyzed the cleavage of lignin aryl methyl bonds, and increased the phenolic groups of lignin through an S_N2 mechanism. The tensile strength of the cured lignin-based adhesives exceeded a petroleum-based epoxy resin adhesive and raw-lignin-based epoxy resin adhesive, and were much higher than the mechanical properties of other epoxy resin adhesives



Scheme 1 Scheme for the demethylation of lignin and preparation of lignin-based epoxy resins (a) and adhesives (b).

prepared in previous work (12.13 MPa[21] and 11.29 MPa[41]). The structure of demethylated lignin was analyzed through FT-IR spectroscopy, 2D HSQC NMR, and GPC. The thermodynamic properties of the adhesive were analyzed by DSC, which can reflect the degree of crosslinking between the lignin component of the adhesive and the epoxy resin. In addition, the tensile properties of aluminum plate lapping with the demethylated lignin adhesive under extreme conditions were tested.

Experimental

Materials

Lignin was purchased from Dongsheng New Material Co., Ltd (Shanghai, China). Lithium bromide (LiBr), hydrobromic acid (HBr) (ACS, 48%), epoxide resin (E-44), epichlorohydrin, *N*,*N*-dimethylformamide (DMF), and *p*-nitrobenzaldehyde were purchased from Macklin Co., Ltd (Shanghai, China). T-31 (curing agent) was purchased from Sanhe Chemical Co, Ltd (Tianjin, China). Sodium hydroxide, hydrochloric acid (AR, 38%), acetone, and acetic anhydride were purchased from Energy Chemical (Shanghai, China). All the deuterated solvents were purchased from Sigma-Aldrich Co, Ltd (USA).

Lignin demethylation approaches in LiBr/HBr systems

First, LiBr was added into 20.00 mL deionized water and stirred until it was totally dissolved. Then, LiBr solution, 3.00 g lignin, and hydrobromic acid were put into a hydrothermal reactor which had a tetrafluoroethylene liner and which was sealed and then heated at 120 °C for a certain time, with stirring at 600 rpm. After heating, the lignin was turned into demethylated lignin (DL). Finally, the reactants were filtrated to gather the solids and washed with deionized water until neutral. The obtained solid was freeze-dried for 48 h. For detailed discussions of the effect of lignin demethylation under different times, temperatures, and LiBr and HBr contents, a series of process condition optimization experiments were carried out. The details are listed in Table S1 (ESI†). The yield of demethylated lignin was calculated by the formula below:

$$Y_{\mathrm{DL}}(\mathrm{wt\%}) = \frac{W_{\mathrm{DL}}}{W_{\mathrm{L}}} \times 100\% \tag{1}$$

where Y_{DL} is the yield of demethylated lignin, and W_{DL} and W_{L} correspond to the weights of demethylated lignin and raw lignin, respectively.

Preparation of lignin epoxy resin adhesive

First, 5.00 g DL-c-2 (see Table S1, ESI†) was dissolved in 100 mL DMF with stirring at 600 rpm for 1 h. Then, 20.00 mL epichlorohydrin and 5.00 mL NaOH aqueous solution (20 wt%) were mixed with DL-c-2 solution, reacted at 80 °C, and stirred at 600 rpm for 3 h in a three-necked flask. After that, the obtained liquid–solid mixture was rotary evaporated to remove epichlorohydrin. The resulting solid-phase products were washed with deionized water several times to remove NaCl and NaOH, and centrifuged at 10 000 rpm for 6 min to obtain the demethylated lignin epoxy resin (DLER). Meanwhile, the raw lignin epoxy resin (RLER) was prepared using raw lignin through the same procedure. Subsequently, the DLER (or RLER), E-44, and 2.46 g T-31 (curing agent) were mixed in the ratio of 4 : 1 of epoxy to T-31, and stirred to a homogeneous state, then cured at 80 °C for 4 h. For a detailed discussion of the performance of the lignin epoxy resin adhesive in the composition of DLER (or RLER) and E-44 with different mass ratios, a series of experiments were carried out. The details are listed in Table S2 (ESI†).

Characterization and measurement

The linkages between the lignin structural units were determined using a Bruker AVIII 400 MHz spectrometer. The 2D HSQC NMR analysis was performed to investigate the differences in the lignin unit linkages between raw lignin and demethylated lignin. The ^{1}H NMR spectra of lignin were obtained to quantitatively calculate the methoxyl content of demethylated lignin and raw lignin according to a previous report.[42] The composition of each lignin ^{1}H NMR sample was: 100.00 mg lignin and 10.00 mg *p*-nitrobenzaldehyde (internal standard) dissolved in 0.50 mL chloroform-d. The calculation formula is as follows:

$$\mathrm{RA} = \frac{I_{\mathrm{OMe}}}{I_{\mathrm{pN}}} \times \frac{M_{\mathrm{pN}}}{M_{\mathrm{L}}} \times 100\% \tag{2}$$

where RA is the relative amount of the methoxyl group content of the lignin samples; I_{OMe} is the integration from 4.05–3.11 ppm, which was assigned to the protons signal of the methoxyl group in the ^{1}H NMR spectra; I_{pN} is the integration from 8.45–8.33 ppm, which was assigned to the protons signal of *p*-nitrobenzaldehyde in the ^{1}H NMR spectra; M_{pN} is the weight of *p*-nitrobenzaldehyde; and M_{L} is the weight of raw lignin.

The hydroxyl groups and carboxyl groups contents of lignin were measured using a 905 Titrando automatic potentiometric titrator. In brief, the lignin samples were dissolved in 0.10 mol L^{-1} KOH solution and titrated by 0.22 mol L^{-1} HCl solution. The formula and other detailed information of the method for calculating the phenolic hydroxyl and carboxyl content were based on the previous work of Zhou *et al.*[43] The molecular weights of demethylated lignin and raw lignin samples were measured by gel permeation chromatography system (GPC, Waters Co., USA) equipped with a RI detector (Waters 2141 Co., USA) and two series columns: TSK-GEL G-4000 PWxl (7.8 × 300 mm) and TSK-GEL G-2500 PWxl (7.8 × 300 mm). The FTIR spectra of the lignin samples were obtained using a Frontier spectrometer (PerkinElmer, USA).

The epoxy values of the demethylated lignin epoxy and raw lignin epoxy were measured through a chemical titration method with reference to the national standard GB/T 1677-2008. SEM analysis of the lignin epoxy adhesive was performed on an Hitachi S-800 system (Hitachi Co., Japan). The thermogravimetric curves of the lignin epoxy adhesive were recorded by a TA Q500 thermal analyzer (TA Instruments, USA) and the procedure was: heat from 25 °C to 700 °C, at a temperature ramp of 10 °C min^{-1} with a N_2 flow rate of 40 mL min^{-1}. A TA discovery DSC 250 analyzer (TA Instruments, Co.) was employed to collect the phase-transition behavior information of the lignin epoxy adhesive, and the procedures was: in the first round, the flange temperature was equilibrated to 0 °C with a 2 min isothermal process, followed by heating from 0 °C to 150 °C at a temperature ramp of 2 °C min^{-1}; the second round was the same as for the same procedure in the first round, except the heating was from 0 °C to 300 °C. The recorded DSC date from the second round were accepted in order to make the date more accurate.

To test the tensile strength of the lignin epoxy adhesive, the adhesive was coated on one side of a 2A12-T4 aluminum alloy plate (the dimensions were 100 × 25 × 2 mm). Before each preparation, the uncoated aluminum alloy plate was polished with 800 mesh sandpaper, immersed in acetone for 24 h, and dried at 105 °C for later use. The coated area was 25 mm × 12 mm, and the two coated plates overlapped exactly within the coated part, and the thickness of the coated adhesive was 200 μm, which was cured at 100 °C for 3 h in an air-drying oven. The tensile strength test procedures referred to national standard GB/T 7124-2008, and an Instron 5965 universal testing machine (Instron Tools Works Co., USA) was used. The space between two collets was 112 mm, and the increase rate of the shear force was 8.3 MPa min^{-1}. To ensure reliable results, the same sample was tested five times. The extreme condition tensile strength test procedure for the lignin-based epoxy resin adhesives was the same as in the mild condition runs except for the different ambient conditions in the anti-water tests and the anti-frozen tests. Five parallel experiments were also performed on the same sample.

Table 1 Detailed parameters of raw lignin and the obtained demethylated lignin, including the molecular weight, dispersity, hydroxyl group content, and demethylated lignin yield

Samples	M_w (g mol^{-1})	M_n (g mol^{-1})	Dispersity index	–OH content (mmol g^{-1})	DL yield (wt%)
Raw Lignin	1526	1099	1.59	3.63	
DL-a-1[a]	2111	1403	1.50	3.59	89.0
DL-a-2[a]	1929	1366	1.31	5.4	89.4
DL-a-3[a]	1498	1218	1.92	6.05	91.2
DL-a-4[a]	1579	1267	1.85	5.91	88.3
DL-b-1[b]	1493	1224	1.92	3.54	90.7
DL-b-2[b]	1453	1197	1.95	5.35	92.0
[g]DL-b-3[b]	1450	1191	1.96	5.25	90.8
DL-c-1[c]	1423	1023	1.39	6.15	90.3
DL-c-2[c]	1313	942	1.47	6.19	91.0
DL-c-3[c]	1632	1102	1.39	5.70	89.7
DL-d-1[d]	1544	1024	1.36	3.34	85.0
DL-d-2[d]	1596	1080	1.35	5.62	87.7
DL-d-3[d]	1679	1114	1.36	4.90	84.3

[a] 5 mL HBr, 30 min, 120 °C, dosage of LiBr: 0, 15, 30, 45 g, respectively. [b] 30 g LiBr, 5 mL HBr, 30 min, temperature: 110, 130, 140 °C, respectively. [c] 30 g LiBr, 5 mL HBr, 120 °C, time: 45, 60, 70 min, respectively. [d] 30 g LiBr, 120 °C, 60 min, HBr: 3, 7, 9 mL, respectively.

Results and discussion

Effect of the LiBr/HBr system on lignin structure

In order to achieve a homogeneous DL with a high phenolic hydroxyl content for the preparation of lignin-based epoxy resin, a series of optimization experiments (reaction conditions are listed in Table S1, ESI†) for the LiBr/HBr system was carried out, and the detailed information is listed in Table 1. As shown in the table, the M_w of DL first decreased and then increased with the increased dosage of HBr. Interestingly, in DL-a-1, which reacted with the free LiBr, the M_w was up to 2111 g mol^{-1}. These results could be attributed to the carbenium ion in the α position of the lignin side chain condensed with the negatively charged benzene ring, with the resulting condensation structure leading to the DL with a high M_w.[44] In addition, the –OH content of DL-a-1 was 3.59 mmol g^{-1} before the reaction, and then raised to 6.05 mmol g^{-1} with the addition of 30 g LiBr in demethylation system, and the hydroxyl content of the DL first increased and then decreased with the increased dosage of LiBr. This phenomenon revealed the significance of LiBr for catalyzing the cleavage of lignin aryl methyl bonds, forming bromomethane and enhancing the hydroxyl group content *via* the S_N2 mechanism.[45] In addition, with the increase in the temperature (from 110 °C to 140 °C), the M_w of DL slightly decreased, and the molecular dispersity of DL stayed around 1.95, which meant that the temperature had little effect on the molecular weight of DL. However, the phenolic hydroxyl content of DL was 3.54 mmol g^{-1} (DL-b-1) after reacting at 110 °C, but then significantly increased to 6.05 mmol g^{-1} at 120 °C (DL-a-3), before finally decreasing to 5.25 mmol g^{-1} at 140 °C (DL-b-3). This phenomenon indicated that the demethylation degree of DL was increased with the increase in the reaction temperature, resulting in the increase in the phenolic hydroxyl content, while the reduction at 140 °C was possibly due to the re-etherfication of the phenolic hydroxyl.[45] In addition, with the increase in the reaction time (from 30 min to 75 min), the M_w first decreased then increased. This could possibly be attributed to the cleavage of the lignin polymer-chains through the extending of the reacting time, whereby the lignin fragments were re-condensed along with the reacting time. With the increase in the reaction time, the phenolic content also first increased and then decreased, which meant that the demethylation degree of DL was growing with the extension of the reaction time, and the re-etherfication also happened at the extended reaction time. In addition, the increased amount of HBr enhanced the depolymerization of DL (from 1544 g mol^{-1} of DL-d-1 to 1313 g mol^{-1} of DL-c-2) and the demethylation degree of DL (from 3.34 mmol g^{-1} of DL-d-1 to 6.19 mmol g^{-1} of DL-c-2). However, excessive HBr also brought the problem of too high a molecular weight and decreased the phenolic hydroxyl content in DL. This was possibly because the excess HBr caused the severe condensation of DL during the demethylation process.[45] Besides, all the recovered DL lignin yields were above 85%, and the treatment time was only 1 h to double the –OH content compared with raw lignin, indicating that the demethylation system was efficient and only needed a short time. Finally, the optimized demethylation condition was concluded to be 3 g raw lignin, 30 g LiBr, 5 mL HBr, and heating at 120 °C for 1 h.

The variation of the lignin methoxyl content in DL after the demethylation process was measured by quantitative ^{1}H NMR analysis based on the previous report of Mancera *et al.*[42] The ^{1}H NMR spectra and relative amounts of methoxyl of raw lignin and DL-c-2 are shown in Fig. S1 (ESI†). The chemical shift ranging from 8.45 to 8.33 ppm was assigned to the aromatic protons peaks of *p*-nitrobenzaldehyde, while the peaks at 4.05–3.11 ppm were assigned to the $-CH_3$ peaks of methoxyl in the lignin samples. In DL-c-2, RA = 0.61, which was significantly lower than that of raw lignin (RA = 0.83), indicating that the LiBr/HBr system efficiently catalyzed the breakage of the aryl methyl ether bond inside the lignin structural units. With the high freedom of H^+ ions in the LiBr/HBr solution, abundant methoxyl was substituted by the high concentration of Br^- through S_N2 substitution.[46] 2D HSQC NMR was also employed to reveal the structural changes during the lignin demethylation process in the LiBr/HBr systems, as shown in Fig. 1(a) (raw lignin) and Fig. 1(b) (DL-c-2). The raw lignin contained abundant inner linkages, including methoxyl (OMe, δC/δH = 56.05/3.72), β-O-4 linkage (A, δC/δH = 72.07/5.06), resinol structure (B, δC/δH = 85.13/4.75), and phenylcoumaran structure (C, δC/δH = 86.22/5.32). Besides, a signal attributed to carbohydrates was observed in the aliphatic areas of raw lignin in the HSQC spectra (Fig. 1(a)). However, most of these linkages were weakened or vanished in the HSQC spectra of DL-c-2. Significantly, the signal for Cα (δC/δH = 72.07/5.06) in the β-O-4 subunit clearly existed in the HSQC spectra of raw lignin, but had totally disappeared in the spectra of DL-c-2. These results revealed that LiBr/HBr possessed the ability for the cleavage of lignin ether bonds followed by the generation of lignin fragments, and with the synergy of H^+ in the demethylation system, an increased amount of phenolic hydroxyl groups in DL-c-2 compared with raw lignin could be predicted. The FTIR spectra of the lignin samples were recorded, as well. Two clear vibrations assigned to –OH and $-CH_3$ (assigned to OMe) at 3431 and 2940 cm^{-1}, respectively, could be observed in raw lignin and demethylated lignin in Fig. S2 (ESI†). Significantly, in the absorption curves of DL-c-2, compared with raw lignin, the vibration of the hydroxyl group grew wider, while the vibration of $-CH_3$ almost disappeared. This indicated that the phenolic group content of DL c-2 was higher than that of raw lignin, and the decrease in the amount of methoxyl group in DL-c-2 was verified.

Effect of lignin on the properties of the adhesive

In order to unlock the mechanism of adhesive curing and cross-linking, the FTIR spectra of the lignin samples, epoxy resin, and adhesive (prepared as described in Sections 2.2 and 2.3) were recorded and analyzed. As shown in Fig. 2(a), the peak at 2940 cm^{-1} was assigned to the $-CH_3$ in the lignin and adhesives samples. As mentioned in Section 3.1, the sharp decline in the methoxyl group content resulted from the demethylation of the lignin in the LiBr/HBr system. However, after the preparation of

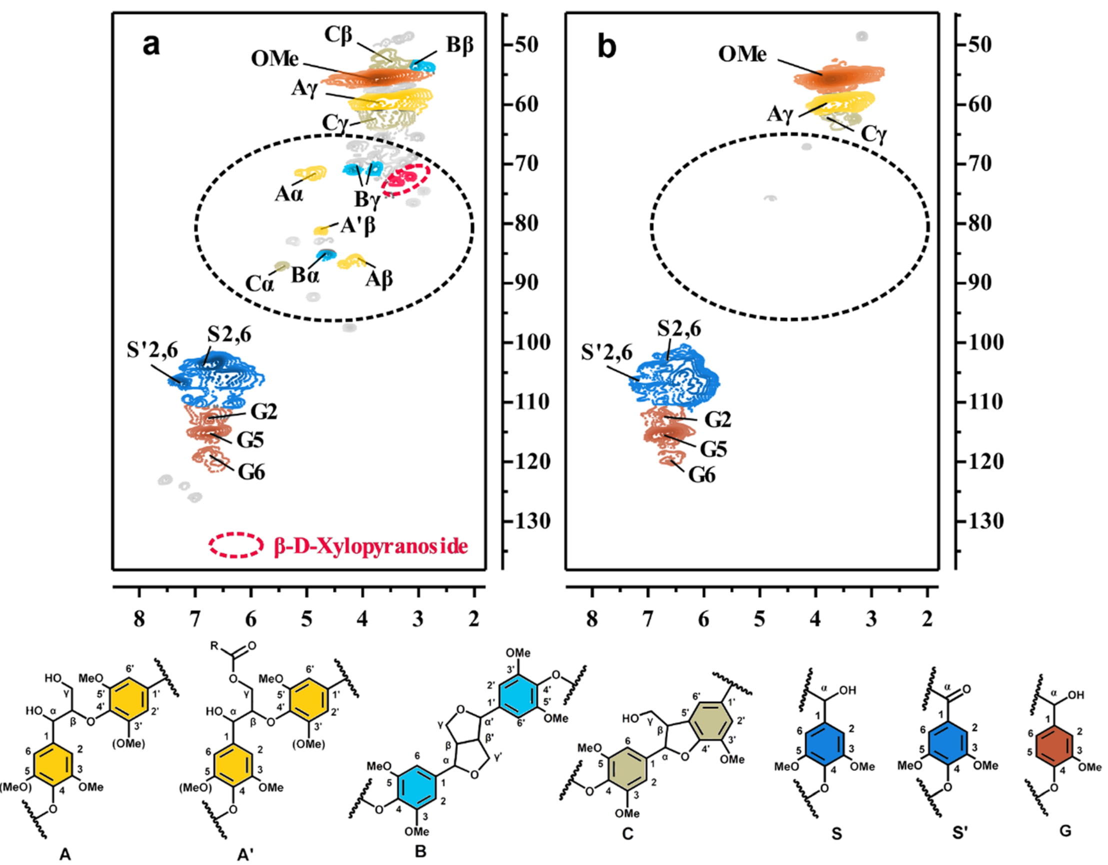


Fig. 1 2D HSQC NMR spectra of raw lignin (a) and DL-c-2 (b), and the main structures: (A) β-O-4 linkages bearing a hydroxyl group at Cγ; (A′) β-O-4 linkages bearing an acetyl group at Cγ; (B) resinols; (C) phenylcoumarans; (G) guaiacyl units; (S′) oxidized syringyl units bearing a carbonyl group at Cα; (S) syringyl units.

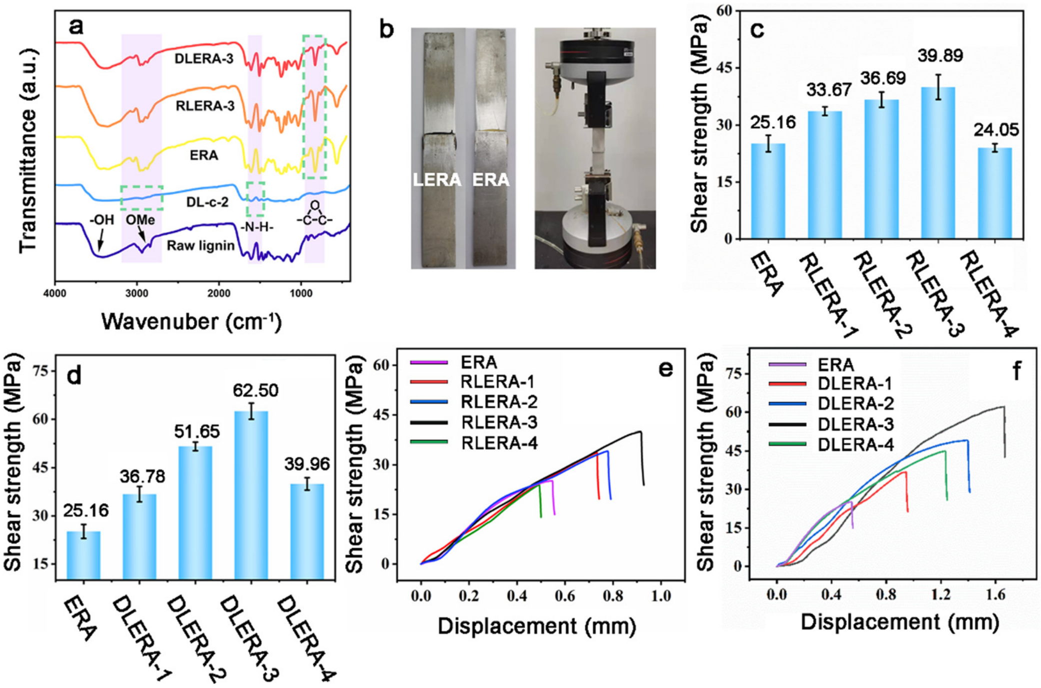


Fig. 2 Structural analysis and mechanical property of LERAs. (a) FTIR spectra of the lignin samples and prepared adhesives; (b) the two overlapping aluminum alloy plates used different LERAs and the adhesion quantification of that was determined by lap shear strength tests shear strengths (c) and tensile stress–strain curves (e) of ERA (as control) and RLERA (different additional amount of RL during the preparation process of the RLERA); shear strengths (d) and tensile stress–strain curves (f) of ERA (as control) and DLERA (different additional amount of DL).

ERA, RLERA-3, and DLERA-3 using DL-c-2, the vibrations of $-CH_3$ and $-CH_2$ reappeared in the FTIR plots. This indicated that the grafting and crosslinking of DL-c-2, epichlorohydrin, E-44, and T-31 was successful. In addition, the peaks ranging from 1640 to 1500 cm^{-1} were attributed to the vibration of the –NH group, which was nonexistent with DL-c-2, but appeared with ERA, RLERA-3, and DLERA-3. These results can be explained by the crosslinking between the end $-NH_2$ group of T-31 and the end epoxy group of the lignin epoxy resin and the formation of a secondary amine.[47] In addition, in contrast to raw lignin and DL-c-2, the occurrence of epoxy group vibrations at 912 and 860 cm^{-1} in the three prepared lignin-based epoxy resin adhesives (LERA) were also proof of the grafting of epichlorohydrin on lignin. In addition, the epoxy value of RLER (0.363) was lower than for DLER (0.454, prepared using DL-c-2), suggesting that DL-c-2 had better reactivity for the epoxidation reaction because of its higher phenolic hydroxyl group content. In short, it was found that a kind of good crosslinking and a possible high molecular weight of lignin epoxy resin adhesive were obtained.

For the evaluation of the strength capacity of LERA, lab shear tests were carried out. The tensile shear strengths of ERA and LERA are shown in Fig. 2(c) and (d). The preparation of ERA only involved E-44 cured and crosslinked by T-31 and the tensile strength of ERA was 25.16 MPa. Compared with ERA, with the introduction of lignin, the tensile shear strength of all the lignin-based LERA showed a significant increase, especially in DLERA-3 (62.50 MPa), which was up to 248.41% (in comparison with ERA with 25.16 MPa). These results were mainly due to the addition of lignin and could improve the tensile shear strength of LERA.[26,47] When lignin was crosslinked with E-44, abundant aromatic ring units were also introduced into the polymer chain of epoxy resin adhesive, and the crosslinking density of LERA was improved; thus the toughness and mechanical properties of the adhesives layer were raised. Besides, the tensile shear strength of the adhesives prepared with the same lignin increased first and then decreased with the increase in lignin content. This phenomenon possibly could only be explained by agglomeration of the lignin. Initially, the low amount of additional lignin in LERA could be well crosslinked with E-44 and T-31, showing an improvement in the mechanical properties on the LERA. However, when the addition amount of lignin exceeded 35%, the agglomerative of lignin particles began, and lots of uncrosslinked lignin simply filled the space between the adhesive polymer networks. When tensile stress was applied to the adhesive, the stress was concentrated caused by the uneven force, and brittle fractures of the material occurred,[27] which indicated a decline in the tensile shear strength of LERA.

The glass-transition temperature (T_g) of LERA can reflect the crosslink density of adhesive composites.[41] The higher the crosslinking density of the molecular chain in the adhesive, the more difficult the free movement of the molecular chain segments, and the higher the T_g of the adhesive. As shown in Fig. 3, in RLERA and DLERA, the T_g of all the adhesives first increased and then decreased with increasing the amount of the additional lignin. These results possibly resulted from the crosslinking density of the adhesives. In the case of a relatively

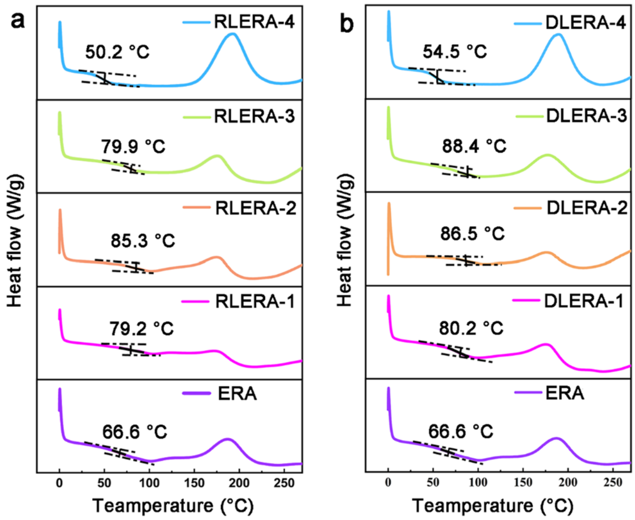


Fig. 3 DSC curves of ERA (as control), RLERAs (a), and DLERAs (b).

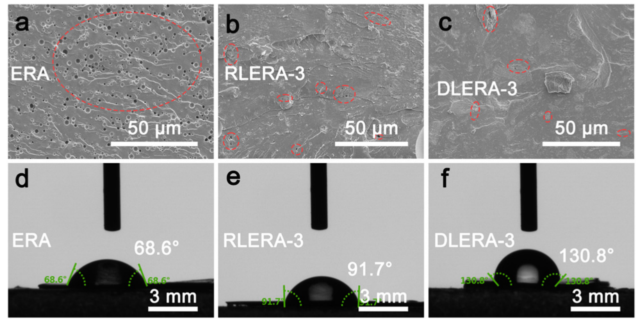


Fig. 4 SEM images of the flexural fracture surfaces of ERA (a), RLERA-3 (b), and DLERA-3 (c). The contact angle of water on ERA (d), RLERA-3 (e), and DLERA-3 (f).

low lignin content in the adhesives, the crosslinking density increased with the addition of more lignin, and more benzene rings were crosslinked in the molecular chain segments in the adhesive, which hindered the movement of the molecular chain segments and led to a rise in the glass-transition temperature of the adhesive. However, excess lignin in the adhesive caused a decline of the crosslinking density of the adhesive, as well as a decline in T_g. These results were in line with the data of the tensile shear strengths of RLERA and DLERA in Fig. 2. Besides, as shown in Fig. 3, when the additional RLER (Table S2, ESI†) was up to 30 wt%, the T_g of RLERA-3 was 79.9 °C, lower than that of RLERA-2 (85.3 °C) and DLERA-3 (88.4 °C). This result indicated that, compared with RL, DL could crosslink with more E-44 to form a more robust LERA network. Interestingly, the T_g of RLERA-2 (20 wt% of the RLER) was 85.3 °C, and the T_g of RLERA-3 decreased to 79.9 °C with increasing the content of RLER (30 w%). However, for DLERA-3, which contained 30 wt% of DLER, the T_g was 88.4 °C, still higher than that for the T_g (86.5 °C) of DLERA-2 (containing 20 wt% of the DLER). This phenomenon indicated that more demethylated lignin could be used to prepare lignin-based epoxy resin adhesives to solve the problem of the decrease in the crosslinking density and mechanical properties when the additional lignin content was too high. Fig. S3 (ESI†) shows the TG and DTG curves of ERA, RLERA, and DLERA, and also detailed information about the thermal stabilities of the adhesives is given. The maximum pyrolysis temperature of ERA was 393.6 °C, higher than for all of these adhesives, indicating that the ERA had the best thermal stability of all the adhesive samples. With the increased amount of additional lignin-based epoxy resin, the pyrolysis temperature of the adhesive declined, and so the adhesive could be easier decomposed.

SEM images of lignin-based epoxy resin adhesives were also obtained for further investigation of the difference in crosslinking density of the LERAs. As shown in Fig. 4, the flexural fracture surface of ERA was full of voids and folds. The number of voids were decreased in RLERA-3 and plenty of agglomerates still existed in RLERA-3, but the flexural fracture surface of DLERA-3 was much more compact and smoother. Compared to ERA, the number of small holes in RLERA-3 and DLERA-3 was significantly decreased (marked with red circles in Fig. 4). This phenomenon could be possibly ascribed to the increase in crosslinking density of the RLERAs and DLERAs, and the excellent compatibility of DL with E-44. When the adhesive was prepared using RL, the poor compatibility of RL and E-44 led to the generation of voids and agglomerates of RL. However, compared with RL, DL had more hydroxyl groups than RL, which meant that more hydroxyl bonds were formed between DL and E-44, thus improving the compatibility of DL during the curing process of the adhesives and the crosslinking density of the prepared adhesives (DLERAs). The water contact angle on the lignin-based epoxy resin adhesives could also reflect the crosslinking density of the adhesives. As shown in Fig. 4, the water contact angle on ERA, RLERA-3, and DLERA gradually increased, which was in line with the determination of the crosslinking density of the lignin-based epoxy resin adhesives, as mentioned in the DSC analysis, indicating that the additional DL improved the crosslinking density and the mechanical property of the adhesive, and thus possibly made the adhesive become more anti-water. In addition, excess additional RL or DL would aggregate inside or on the surface of the adhesive, and would thus expose plenty of uncrosslinked hydroxyl groups, thus decreasing the water contact angle on the corresponding adhesive (Fig. S4, ESI†).

In view of the ambient conditions in which the epoxy resin adhesives were used, possibly including high humidity and low temperature, the waterproofing and anti-freezing properties of the prepared adhesives were tested. As shown in Fig. 5, after bathing in the water at 25 °C for 48 h and freezing at −25 °C for 48 h, the tensile shear strengths of all the lignin-based epoxy resins slightly decreased. In order to evaluate the mechanical property loss after the extreme ambient condition tests, the maintenance of the property rates of the ERA, RLERAs, and DLERAs were determined, as shown in Fig. S5 (ESI†). The maintained property rates of these prepared adhesives ranged from 75% to 88%, which exhibited excellent the anti-water and anti-freezing capacities of the prepared lignin-based epoxy resin adhesives. In addition, the maintenance rates of DLERA-1 after

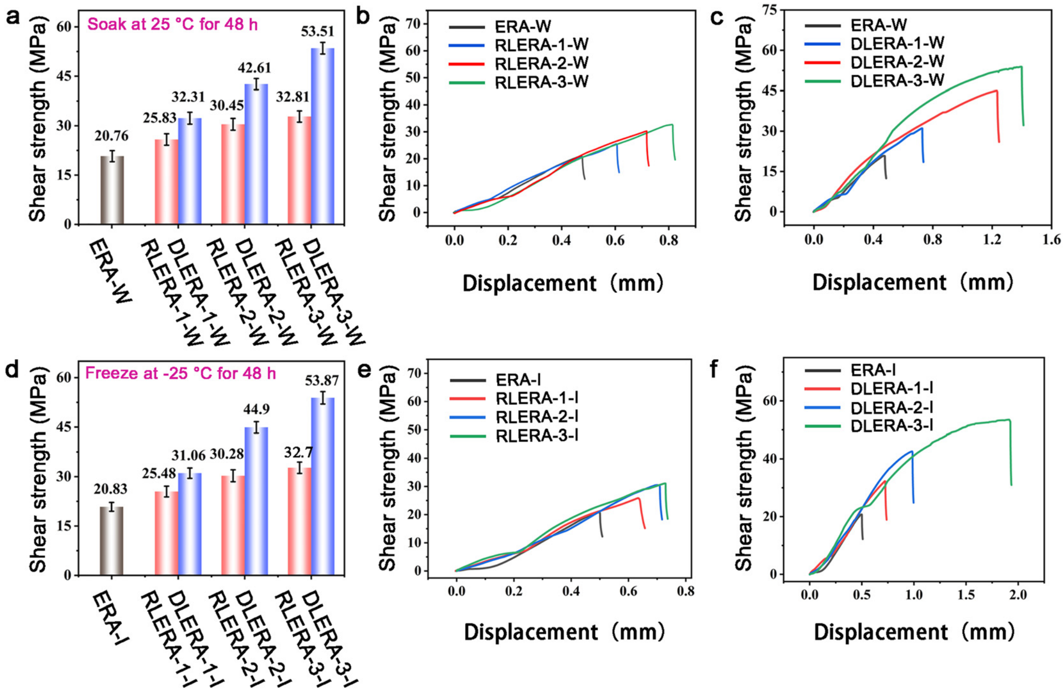


Fig. 5 Mechanical properties of the lignin-based epoxy resin adhesives under extreme conditions. (a) Shear strengths of ERA, RLERA, and DLERA after bathing in water at 25 °C for 48 h; tensile stress–strain curves of ERA (as control), RLERA (b), and DLERA (c) after bathing in water at 25 °C for 48 h; (d) Shear strengths of ERA, RLERA, and DLERA after freezing at −25 °C for 48 h; Tensile stress–strain curves of ERA (as control), RLERA (e), and DLERA (f) after freezing at −25 °C for 48 h.

soaking and freezing were 87.85% and 84.44%, which were higher than RLERA-1 after soaking (76.72%) and freezing (75.68%). This phenomenon indicated that the additional DL lignin could better improve the anti-water and anti-freezing properties than RL, which could possibly be ascribed to the higher crosslinking density in the DLERAs. The increased crosslinking density in DLERAs led to a decrease in the voids and aggregated lignin, and it becoming harder for water to cause swelling in the adhesives, thus improving the anti-water and anti-freezing properties of the adhesives.

Conclusion

In summary, a lignin-based epoxy resin adhesive was prepared by using demethylated lignin. The phenolic hydroxyl content (6.19 mmol g^{-1}) of DL obtained by demethylation in the LiBr/HBr system was 170.52% higher than that of RL, which indicated the greatly improved crosslinking density and tensile shear strength of the demethylated-lignin-based epoxy resin adhesive. When 30% demethylated lignin epoxy resin was used in the preparation of DLERA-3, the tensile shear strength reached 62.50 MPa at room temperature, which was 2.48 times that of a petroleum-based epoxy resin adhesive and 1.57 times of that of RLERA-3. After soaking in water for 48 h or frozen at −25 °C for 48 h, it could retained more than 85% tensile shear strength. The tensile strength of 62.50 MPa indicated that it has great application potential in furniture manufacturing and for improving the performance of plywood. Its excellent water and frost resistance means that furniture or plywood bonded with it should maintain good strength and performance in a variety of scenarios, such as harsh outdoor conditions.

Conflicts of interest

The authors declare no competing financial interest.

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